

A Short Guide to Lemma C.2

Følner averaging, compactness, and exact covariance

Abstract

Lemma C.2 turns an optimization over arbitrary hybrid channels, averaged on larger and larger parameter cubes, into an optimization at the origin over exactly covariant maps. The proof combines four ideas: group averaging, the Følner property of Euclidean cubes, weak-* compactness of channel adjoints, and upper semicontinuity of fidelity. This note develops those ideas and explains why the limiting map may lose trace when mass escapes in the noncompact classical coordinate.

1 What the lemma accomplishes

The hybrid Gaussian experiment has a classical coordinate $h \in \mathbf{H}$ and a quantum displacement coordinate $z \in \mathbb{C}^{r_d}$. Write

$$\mathbf{G} = \mathbf{H} \times \mathbb{C}^{r_d}, \quad \xi = (h, z).$$

The input and desired target form two covariant families

$$\Theta_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{in}}(\Theta_0^{\text{in}}), \quad \Theta_\xi^{\text{tar}} = \mathcal{U}_\xi^{\text{out}}(\Theta_0^{\text{tar}}).$$

For a channel Λ , the paper studies the average payoff

$$\frac{1}{|\Omega_L|} \int_{\Omega_L} \mathbf{B}(\Lambda(\Theta_\xi^{\text{in}}), \Theta_\xi^{\text{tar}}) \, d\xi$$

over a large centered cube $\Omega_L \subset \mathbf{G}$. An arbitrary channel can behave differently at different parameter values. Lemma C.2 says that its asymptotic average performance cannot exceed the performance at the origin of some map Γ satisfying the exact intertwining relation

$$\Gamma \mathcal{U}_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{out}} \Gamma \quad (\xi \in \mathbf{G}).$$

Thus a noncompact, many-parameter optimization is reduced to a single-state optimization with a strong symmetry constraint.

2 The hybrid state spaces

Let $\mathcal{F}_{\text{orb}}$ be the multimode Fock space. An ordinary hybrid state is a Bochner-integrable field of trace-class operators,

$$\mathfrak{T} = L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})).$$

Its natural test algebra is

$$\mathfrak{A} = C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}})), \quad \mathfrak{M} = \mathfrak{A}^*.$$

Positive elements of $\mathfrak{M}$ are finite positive $\mathcal{T}_1(\mathcal{F}_{\text{orb}})$ -valued Radon measures. Allowing outputs in $\mathfrak{M}$, rather than only in $\mathfrak{T}$, is important: weak limits may have singular classical parts such as point masses.

The group actions used in the paper are

$$\begin{aligned}(\mathcal{U}_\xi^{\text{in}} R)(x) &= D(z)R(x-h)D(z)^*, \\ (\mathcal{U}_\xi^{\text{out}} \Xi)(E) &= D(\sqrt{\gamma}z) \Xi(E - \sqrt{\gamma}h) D(\sqrt{\gamma}z)^*.\end{aligned}$$

They translate the classical component and conjugate the quantum component by Weyl displacements. Both preserve positivity and total trace. Root fidelity is invariant under these reversible actions.

Since $\mathfrak{A}$ is nonunital, the proof also uses its unitization

$$\tilde{\mathfrak{A}} = \mathfrak{A} \oplus \mathbb{C}I.$$

A normalized hybrid measure has a canonical state extension to $\tilde{\mathfrak{A}}$. The extra quotient character of $\tilde{\mathfrak{A}}$ represents a point at infinity. This compactifies the state space sufficiently for weak-* extraction.

3 Background idea I: Følner sets

A locally compact group is amenable if it possesses sets whose relative boundary becomes negligible under every fixed translation. Here the group is a Euclidean vector group, and centered cubes form a Følner family:

$$\frac{|(\Omega_L + v) \triangle \Omega_L|}{|\Omega_L|} \longrightarrow 0 \quad (L \rightarrow \infty) \quad (1)$$

for every fixed $v \in \mathbf{G}$. The symbol $\triangle$ denotes symmetric difference.

Geometrically, translation by a fixed vector changes only slabs near the faces of a large cube. Their volume is of order $L^{\dim \mathbf{G}-1}$, whereas the cube has volume of order $L^{\dim \mathbf{G}}$. Their ratio therefore tends to zero. Equation (1) is the mechanism that turns approximate covariance of averaged channels into exact covariance of a limit.

4 Background idea II: compactness through adjoints

There is no useful norm compactness for a sequence of infinite-dimensional channels. Instead one passes to adjoints. A trace-preserving completely positive map induces a unital completely positive contraction

$$\hat{\Lambda}^* : \tilde{\mathfrak{A}} \longrightarrow \mathfrak{T}^*, \quad \mathfrak{T}^* = L^\infty(\mathbf{H}; \mathcal{B}(\mathcal{F}_{\text{orb}})).$$

The unit balls of the dual space $\mathfrak{T}^*$ are weak-* compact. Because $\tilde{\mathfrak{A}}$ and the predual $\mathfrak{T}$ are separable, weak-* compactness is metrizable on bounded subsets. One may therefore choose a countable dense operator system in $\tilde{\mathfrak{A}}$ and diagonalize to obtain pointwise weak-* convergence of the adjoints.

Complete positivity survives this limit. Indeed, for every matrix size r , positivity of a map means

$$[A_{kl}] \geq 0 \implies [\hat{\Lambda}^*(A_{kl})] \geq 0.$$

The positive cone of the dual matrix space is weak-* closed, so this property passes to the limiting adjoint at every r . This is why the proof checks matrix positivity, rather than only ordinary positivity.

5 Statement of Lemma C.2

Lemma C.2 (rephrased). *Let Ω_L be the centered cubes in $\mathbf{G}$. Then*

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} B(\Lambda(\Theta_\xi^{\text{in}}), \Theta_\xi^{\text{tar}}) d\xi \leq \sup_{\Gamma} B(\Gamma(\Theta_0^{\text{in}}), \Theta_0^{\text{tar}}), \quad (2)$$

where Λ ranges over hybrid channels and Γ ranges over completely positive, trace-nonincreasing maps $\mathfrak{T} \rightarrow \mathfrak{M}$ such that

$$\Gamma \mathcal{U}_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{out}} \Gamma \quad (\xi \in \mathbf{G}).$$

The output of Γ is allowed to be a measure rather than an ordinary L^1 density.

The direction of (2) matters. The lemma supplies an upper bound: every nearly optimal sequence of arbitrary channels produces an exactly covariant limiting competitor whose origin fidelity is at least the limiting average payoff.

6 Proof, step by step

6.1 Step 1: average the channel in a moving frame

Define

$$\bar{\Lambda}_L = \frac{1}{|\Omega_L|} \int_{\Omega_L} \mathcal{U}_{-\xi}^{\text{out}} \circ \Lambda \circ \mathcal{U}_\xi^{\text{in}} d\xi. \quad (3)$$

For a parameter ξ , the input is first shifted from the origin to ξ , then processed by Λ , and finally shifted back in the output coordinates. Thus every parameter point is viewed from the origin before the results are averaged.

Because $\Theta_\xi^{\text{in}} = \mathcal{U}_\xi^{\text{in}}(\Theta_0^{\text{in}})$ and $\Theta_\xi^{\text{tar}} = \mathcal{U}_\xi^{\text{out}}(\Theta_0^{\text{tar}})$, invariance of root fidelity gives

$$B(\mathcal{U}_{-\xi}^{\text{out}} \Lambda(\Theta_\xi^{\text{in}}), \Theta_0^{\text{tar}}) = B(\Lambda(\Theta_\xi^{\text{in}}), \Theta_\xi^{\text{tar}}).$$

Root fidelity is jointly concave, hence concave in its first argument with the target fixed. Applying this to (3) yields

$$\frac{1}{|\Omega_L|} \int_{\Omega_L} B(\Lambda(\Theta_\xi^{\text{in}}), \Theta_\xi^{\text{tar}}) d\xi \leq B(\bar{\Lambda}_L(\Theta_0^{\text{in}}), \Theta_0^{\text{tar}}). \quad (4)$$

This is the first decisive reduction: an average of fidelities is bounded by one fidelity at the origin.

6.2 Step 2: extract a weak-* limiting map

Choose $L_j \rightarrow \infty$ and channels whose left side in (2) approaches the limsup. For brevity, use $\bar{\Lambda}_j$ for their averages. The unitized adjoints

$$\hat{\Lambda}_j^* : \tilde{\mathfrak{A}} \rightarrow \mathfrak{T}^*$$

are unital completely positive contractions. The diagonal compactness argument described above gives, after passing to a subsequence,

$$\hat{\Lambda}_j^*(A) \xrightarrow{w^*} \hat{\Gamma}^*(A) \quad (A \in \tilde{\mathfrak{A}}).$$

Contractivity first defines the limit on a countable dense operator system and then extends it uniquely to all of $\tilde{\mathfrak{A}}$. Weak-* closedness of every matrix positive cone makes $\hat{\Gamma}^*$ completely positive, and convergence at I makes it unital.

Restrict $\hat{\Gamma}^*$ to the ideal $\mathfrak{A}$ and define $\Gamma : \mathfrak{T} \rightarrow \mathfrak{M}$ by

$$\langle \Gamma(R), A \rangle = \langle R, \hat{\Gamma}^*(A) \rangle, \quad A \in \mathfrak{A}. \quad (5)$$

This restriction is the measure-valued limiting map needed on the right side of (2).

6.3 Step 3: explain why only trace nonincrease remains

Let (e_ℓ) be an increasing contractive approximate identity for $\mathfrak{A}$. For $R \geq 0$, the total mass of the positive measure $\Gamma(R)$ is

$$\mathrm{Tr} \Gamma(R) = \sup_{\ell} \langle \Gamma(R), e_\ell \rangle.$$

Using (5), positivity, and $0 \leq e_\ell \leq I$ gives

$$\begin{aligned} \mathrm{Tr} \Gamma(R) &= \sup_{\ell} \langle R, \hat{\Gamma}^*(e_\ell) \rangle \\ &\leq \langle R, \hat{\Gamma}^*(I) \rangle = \mathrm{Tr} R. \end{aligned}$$

Thus Γ is trace nonincreasing.

The apparent loss is not a defect in the argument. The unitized limit is still normalized, but part of its classical mass may sit at the quotient point at infinity. Restricting from $\tilde{\mathfrak{A}}$ to $\mathfrak{A}$ discards precisely that part. Requiring the restricted limit to remain trace preserving would incorrectly assume tightness of all output measures.

6.4 Step 4: the limiting map is exactly covariant

For a fixed $v \in \mathbf{G}$, translating the integration variable in (3) shows that the two maps $\bar{\Lambda}_j \mathcal{U}_v^{\mathrm{in}}$ and $\mathcal{U}_v^{\mathrm{out}} \bar{\Lambda}_j$ differ only on the symmetric difference of the two averaging cubes. Consequently, for $R \in \mathfrak{T}$,

$$\left\| \bar{\Lambda}_j(\mathcal{U}_v^{\mathrm{in}} R) - \mathcal{U}_v^{\mathrm{out}} \bar{\Lambda}_j(R) \right\|_{\mathfrak{M}} \leq \frac{|(\Omega_{L_j} + v) \triangle \Omega_{L_j}|}{|\Omega_{L_j}|} \|R\|_1. \quad (6)$$

The Følner ratio tends to zero by (1). Pairing (6) with $A \in \mathfrak{A}$ and passing to the pointwise weak-* limit gives

$$\Gamma \mathcal{U}_v^{\mathrm{in}} R = \mathcal{U}_v^{\mathrm{out}} \Gamma(R).$$

Since v was arbitrary, Γ is exactly covariant under all of $\mathbf{G}$.

6.5 Step 5: pass the fidelity payoff to the limit

Weak-* convergence does not generally imply convergence of fidelity. The needed one-sided statement follows from the Alberti–Uhlmann variational formula on a unital C^* -algebra:

$$\mathrm{B}(\phi, \psi) = \inf_{A > 0} \inf_{\text{invertible}} \frac{1}{2} (\phi(A) + \psi(A^{-1})). \quad (7)$$

For fixed ψ , each expression inside the infimum is weak-* continuous in ϕ . An infimum of continuous functions is upper semicontinuous, so

$$\phi_j \xrightarrow{w^*} \phi \implies \limsup_j \mathrm{B}(\phi_j, \psi) \leq \mathrm{B}(\phi, \psi).$$

Apply this on $\tilde{\mathfrak{A}}$ to the unitized origin outputs. Lemma C.1 then removes any mass at infinity: the target is normalized on $\mathfrak{A}$, so its overlap with the quotient component is zero. Therefore

$$\limsup_j \mathrm{B}(\bar{\Lambda}_j(\Theta_0^{\mathrm{in}}), \Theta_0^{\mathrm{tar}}) \leq \mathrm{B}(\Gamma(\Theta_0^{\mathrm{in}}), \Theta_0^{\mathrm{tar}}).$$

Combining this inequality with (4) proves (2).

7 A finite-dimensional analogy

For a finite group G , one may average a channel exactly:

$$\bar{\Lambda} = \frac{1}{|G|} \sum_{g \in G} \mathcal{U}_{g^{-1}}^{\text{out}} \Lambda_g^{\text{in}}.$$

This average is already exactly covariant, so no limit or compactness argument is required. Lemma C.2 is the noncompact analogue. A large Følner cube plays the role of an approximate uniform distribution on G , and weak-* compactness supplies a limiting average because no finite translation-invariant probability measure exists on G .

8 Dependencies and role in the paper

Lemma C.2 uses:

- joint concavity and invariance of root fidelity;
- the Følner property of Euclidean cubes;
- weak-* compactness and separability to extract channel adjoints;
- weak-* closedness of matrix positive cones;
- the Alberti–Uhlmann variational formula; and
- Lemma C.1 to discard fidelity-invisible mass at infinity.

Its output is the starting point for Lemma C.3 and Proposition C.4. Lemma C.3 describes the structure of an exactly covariant hybrid map through a residual subinstrument. Proposition C.4 then bounds the origin fidelity of every such map by

$$B_{\text{univ}}(\gamma, p) = B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

Together, C.2 and C.4 give the upper bound in the hybrid Gaussian flat-prior theorem.

9 Minimal prerequisite checklist

The conceptual core requires only covariance, concavity of root fidelity, and the fact that the boundary-to-volume ratio of a large cube vanishes. For the rigorous compactness step, one additionally needs dual Banach spaces, the weak-* topology, adjoints of completely positive maps, unitization of a nonunital C^* -algebra, and the variational characterization (7). The main message is simple: average a channel in the moving frame, take a compact limit, and use vanishing boundary effects to make the limit exactly symmetric.